HTML QUICK-STYLER

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ARUNACHALA ARTS & SCIENCE (WOMEN) COLLEGE

Submitted to the

In partial fulfilment of the requirement

for the award of the degree of

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

MANONMANIAM SUNDARANAR UNIVERSITY

TIRUNELVELI -627012

MARCH -2026

Introduction:

HTML Quick Styler is a web-based project that helps users quickly create and improve the visual design of HTML webpages. Many beginners in web development know how to write HTML content, but they often struggle to design attractive layouts, choose good colors, and apply proper styling using CSS. This project solves that problem by automatically generating styling suggestions and design elements for a webpage.

The main goal of this project is to make webpage styling **faster, easier, and more visually appealing**. Instead of manually writing large amounts of CSS code, the system can generate styles, color palettes, and layout improvements automatically. This helps beginners, students, and developers create professional-looking webpages without needing advanced design skills.

The project uses **Python programming** along with modern libraries such as **Gradio** and **Transformers**. The Transformers library allows the system to use an **AI text generation model** that can understand user requests and generate useful responses. For example, a user may ask for a modern webpage style or a vibrant theme, and the system can generate suitable design ideas.

Another important part of the project is the **Color Palette Generator**. Choosing good color combinations is one of the most difficult parts of designing a website. The system solves this by automatically generating cohesive color palettes such as modern, vibrant, or ocean themes. Each palette contains colors for different parts of the webpage, including the background, text, borders, and accents.

The project also uses **Gradio**, which creates an easy-to-use web interface. Through this interface, users can interact with the system, provide input, and receive styled HTML suggestions instantly. This makes the project interactive and user-friendly.

Overall, **HTML Quick Styler** combines web design principles with artificial intelligence to simplify the process of styling webpages. It helps users save time, improves creativity in design, and allows beginners to build attractive websites with less effort. This project demonstrates how AI tools can be integrated with web development to make design tasks easier and more efficient.

Project Description

HTML Quick Styler is a web-based tool designed to help users quickly generate styles and improve the appearance of HTML webpages. Many beginners in web development know how to create webpage structure using HTML, but they often find it difficult to design attractive layouts using CSS. This project helps solve that problem by automatically generating styling suggestions and color themes for webpages.

The system allows users to input a request or choose a theme for their webpage. Based on the input, the tool generates suitable styling elements such as **background colors, text styles, layout suggestions, and color palettes**. These styles help improve the overall look and feel of the webpage without requiring the user to manually write complex CSS code.

The project is developed using **Python** and uses several important technologies. It uses the **Transformers library** to generate intelligent suggestions for styling and design. This allows the system to understand user requests and produce useful outputs. The project also uses **Gradio** to create a simple and interactive web interface where users can easily interact with the tool.

One of the main features of the project is the **Color Palette Generator**. This feature creates different color combinations based on themes such as modern, vibrant, ocean, or sunset. These palettes help users choose colors that match well together and improve the visual design of their webpages.

The tool works by processing the user input, generating design suggestions using the AI model, and then displaying the results through the web interface. Users can then apply these styling suggestions to their HTML files to create more visually appealing websites.

Overall, **HTML Quick Styler** makes webpage design easier and faster. It helps beginners learn about styling while also helping developers save time when creating webpage designs. The project demonstrates how artificial intelligence can be used to assist in web development and improve the design process.

Objectives :

1. **To simplify webpage styling**

   The project aims to make it easier for users to style HTML webpages without needing advanced knowledge of CSS. It helps beginners quickly improve the design of their webpages.

2. **To generate automatic design suggestions**

   The system provides styling suggestions such as layout improvements, text styles, and color combinations based on user input.

3. **To create attractive color palettes**

   The project includes a color palette generator that produces different themes such as modern, vibrant, ocean, and sunset to help users choose suitable colors for their webpages.

4. **To reduce the time required for designing webpages**

   Instead of manually writing a lot of CSS code, users can use the tool to quickly generate styles and apply them to their webpages.

5. **To provide an easy-to-use interface**

   The project uses a simple web interface where users can easily enter their requests and get styling suggestions.

6. **To demonstrate the use of AI in web development**

   The project shows how artificial intelligence can help developers generate design ideas and improve the web development process.

7. **To support learning for beginners**

   The tool helps students and beginners understand how different styles and color combinations affect webpage design.

Applications

1. **Web Development Learning**

   This tool is very useful for students and beginners who are learning HTML and CSS. It helps them understand how styling works and how different colors and layouts can improve a webpage.

2. **Quick Website Prototyping**

   Developers can use the tool to quickly create styled webpage designs when building prototypes or testing new website ideas.

3. **Educational Projects**

   The project can be used in schools and colleges for learning purposes. Students can use it to complete web development assignments and projects.

4. **Small Business Websites**

   Small businesses that need simple websites can use this tool to quickly generate attractive webpage styles without hiring professional designers.

5. **Personal Websites and Portfolios**

   Individuals who want to create personal blogs, portfolios, or simple websites can use the tool to design their webpages easily.

6. **UI Design Assistance**

   The tool helps users experiment with different color themes and styling ideas, which can support the design of user interfaces.

7. **Web Design Practice**

   Designers and developers can use the tool to explore different design combinations and improve their creativity in webpage styling.

System Architecture:

The **system architecture** of the **HTML Quick Styler** project explains how different components of the system work together to generate styling suggestions and display them to

the user. The system is designed in a simple structure where the user interacts with the interface, the request is processed by the backend, and the results are generated and displayed.

1. User Interface (Frontend)

The frontend is the part of the system that the user interacts with. In this project, a simple web interface is created using **Gradio**. Through this interface, users can:

- Enter styling requests or themes
- Select design options
- View generated styling suggestions and color palettes

This interface makes the system easy to use even for beginners.

2. Input Processing Module

When the user enters a request, the system first processes the input. This module reads and understands the user's request such as:

- Theme selection (modern, vibrant, ocean, etc.)
- Style preference
- Design suggestions

The processed input is then sent to the styling generation module.

3. Styling Generation Module

This is the main part of the system where styling suggestions are created. The project uses **Python** and the **Transformers library** to generate intelligent responses based on the user input.

The system can generate:

- HTML styling suggestions

- CSS design ideas
- Layout improvements

This module helps automate the styling process.

4. Color Palette Generator

This module generates suitable color combinations for the webpage. Each palette includes colors for:

- Background
- Text
- Buttons or highlights
- Borders or accents

These color palettes help create visually balanced webpage designs.

5. Output Display Module

After the styles and color palettes are generated, the results are sent back to the frontend interface. The user can view:

- Styling suggestions
- Generated design ideas
- Color palette themes

This allows users to apply the suggested styles to their HTML webpages.

6. Backend Processing

The backend is built using **Python**, which connects all modules together. It manages user input, runs the styling generation logic, and sends results to the interface.

Software Requirements:

The **software requirements** for the **HTML Quick Styler** project are the programs and tools needed to develop and run the project.

1. Programming Language

- **Python**
  Python is used as the main programming language to build the project and implement the styling generation system.

2. Libraries and Frameworks

- **Gradio** – Used to create a simple and interactive web interface for users.
- **Transformers** – Used to generate intelligent styling suggestions.
- **Random Library** – Used to generate different color palettes and themes.

3. Development Environment

- **Visual Studio Code (VS Code)** or **any Python IDE** This
  is used to write, edit, and run the project code.

4. Operating System

The project can run on different operating systems such as:

- Windows
- Linux
- macOS

5. Web Browser

A web browser is required to open and use the interface, such as:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

6. Python Version

- Python **3.8 or higher** is recommended to run the project properly.

Hardware Requirements:

The **hardware requirements** for the **HTML Quick Styler** project are the basic physical components needed to run the system.

1. Processor

- **Intel Core i3 or higher** (or equivalent processor)

2. RAM

- **Minimum 4 GB RAM**
  (8 GB RAM recommended for better performance)

3. Storage

- **At least 10 GB of free disk space** to store the project files and required software.

4. Display

- A **monitor or laptop screen** to view and interact with the web interface.

5. Input Devices

- **Keyboard and Mouse** for entering input and controlling the system.

6. Internet Connection

- An **internet connection** may be required to install libraries and download required packages.

These hardware components are sufficient to **develop, run, and test the HTML Quick Styler project smoothly.**

Project Workflow:

The **project workflow** of the **HTML Quick Styler** explains the step-by-step process of how the system works from user input to the final output.

Step 1: User Input

The user opens the web interface and enters a styling request or selects a theme for the webpage.

For example, the user may choose themes like **modern, vibrant, ocean**, etc.

Step 2: Input Processing

The system reads and processes the user's request.
It understands the theme, style preference, and other design requirements entered by the user.

Step 3: Style Generation

After processing the input, the system generates styling suggestions.
Using **Python and the Transformers library**, it creates ideas for:

- HTML styling
- CSS design
- Layout improvements

Step 4: Color Palette Generation

The system then generates suitable **color combinations** based on the selected theme.

These colors can be used for:

- Background
- Text
- Buttons
- Borders

Step 5: Output Generation
The system prepares the final results, including styling suggestions and color palettes.

Step 6: Display Results

The generated styles and color themes are displayed on the **Gradio web interface** so the user can view them easily.

Step 7: Apply the Styles

The user can take the suggested styles and apply them to their **HTML webpage** to improve its appearance.

Features :

1. **Automatic Styling Suggestions**

   The system automatically gives styling ideas for HTML webpages.

2. **Color Palette Generator**

   It generates different color combinations to make webpages look attractive.

3. **Easy User Interface**

   The project has a simple interface where users can easily enter their requests.

4. **Theme-Based Design**

   Users can choose themes like modern, vibrant, or ocean to generate styles.

5. **Fast Design Generation**

   The system quickly creates styling suggestions.

6. **Beginner Friendly**

   It helps beginners learn how to style webpages.

7. **AI-Based Suggestions**

   The project uses AI to generate design and styling ideas.

8. **Improves Webpage Look**

   It helps make webpages more colorful and visually appealing.

Future Enhancements:

1. **More Design Themes**

   More themes and style options can be added to give users more design choices.
2. **Live Preview Feature**

   Users can see a live preview of the styled webpage before applying the design.

3. **Download Styled Code**

   The system can allow users to download the generated HTML and CSS code.

4. **More Color Palette Options**

   Additional color palettes can be added for better design combinations.

5. **Mobile-Friendly Designs**

   The tool can be improved to generate styles that work better on mobile devices.

6. **Advanced AI Suggestions**

   The AI can be improved to give more creative and accurate styling ideas.

7. **User Customization**

   Users can be allowed to edit and customize the generated styles according to their needs.

Conclusion:

The **HTML Quick Styler** project is designed to help users easily improve the design and appearance of HTML webpages. It provides styling suggestions and color palettes that make webpages look more attractive and organized.

This project uses **Python, Gradio, and AI-based tools** to generate design ideas based on user input. The simple interface allows users to enter their requests and quickly receive styling suggestions.

The system is especially useful for **students, beginners, and web developers** who want to create better webpage designs without writing complex CSS code. It saves time and helps users learn how different styles and colors can improve a webpage.

Overall, the **HTML Quick Styler** makes webpage styling easier, faster, and more creative. It also shows how AI can be used to support web development and design.